

Last name _____

First name _____

LARSON—MATH 756—CLASSROOM WORKSHEET 12
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Review

1. Why is $\alpha_f \geq \frac{n}{2}$ for any graph?
2. What other constraints could we add to our IP (and LP) models?
3. What is Balinski's Lemma?
4. Why is Balinski's Lemma true?

New

1. Why does $\alpha = \alpha_f$ for bipartite graphs?
2. What graph classes generalize KE graphs? Ideally we'd like a larger class where (1) it is possible to efficiently recognize that a graph is in the class, and (2) where α can be computed efficiently.
3. What is an *orthogonal representation* of a graph?

4. Find an orthogonal representation for K_3 .

5. Find an orthogonal representation for C_4 .

6. What is the definition of ϑ ?

7. Find $\vartheta(K_3)$.

8. Find $\vartheta(C_4)$.

9. Why is ϑ well-defined?

10. Show: for any graph, $\alpha \leq \vartheta$.